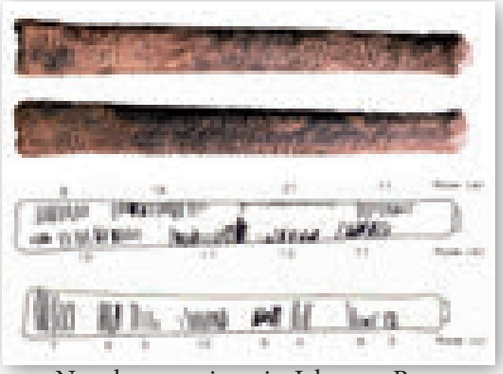
2.1 Introduction

The study of numbers has fascinated humans since several thousands of years. The discovery of Lebombo and Ishango bones which existed around 25000 years ago has confirmed the fact that humans made counting process for meeting various day to day needs. By making notches in the bones they carried out counting efficiently. Most consider that these bones were used as lunar calendar for knowing the phases of moon thereby understanding the seasons. Thus the bones were considered to be the ancient tools for counting. We have come a long way since this primitive counting method existed.



Number carvings in Ishango Bone
Fig.2.1

It is very true that the patterns exhibited by numbers have fascinated almost all professional mathematicians' right from the time of Pythagoras to current time. We will be discussing significant concepts provided by Euclid and continue our journey of studying Modular Arithmetic and knowing about Sequences and Finite Series. These ideas are most fundamental to your progress in mathematics for upcoming classes. It is time for us to begin our journey to understand the most fascinating part of mathematics, namely, the study of numbers.

2.2 Euclid's Division Lemma

Euclid, one of the most important mathematicians wrote an important book named "Elements" in 13 volumes. The first six volumes were devoted to Geometry and for this reason, Euclid is called the "Father of Geometry". But in the next few volumes, he made fundamental contributions to understand the properties of numbers. One among them is the "Euclid's Divison Lemma". This is a simplified version of the long division process that you were performing for division of numbers in earlier classes.

Le us now discuss Euclid's Lemma and its application through an Algorithm termed as "Euclid's Division Algorithm".

Lemma is an auxiliary result used for proving an important theorem. It is usually considered as a mini theorem.

Theorem 1: Euclid's Division Lemma

Let a and b be any two positive integers. Then, there exist unique integers q and r such that $a = bq + r,\ 0 \leq r < b$.

Note

- The remainder is always less than the divisor.
- If $r = 0$ then $a = bq$ so b divides a.
- Conversely, if b divides a then $a = bq$

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